

Session: 2018-2019

Semester: Winter Semester

Name of Examination: 1st Year B. Tech (Common) End Semester Examination

Subject: Electronics Engineering

Subject Code: ECC12101

Max Time: 3 Hours

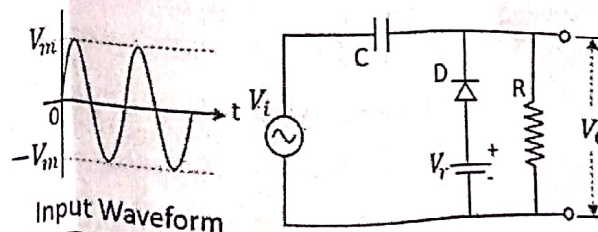
Full Marks - 100

Instructions:

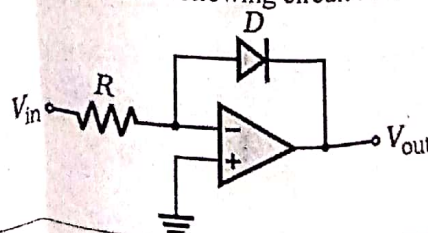
1. Numbers in the rightmost columns represent marks of the corresponding questions.
2. Answer *all* the questions from the Group 1 and *any* 3 from the group 2

Group 1

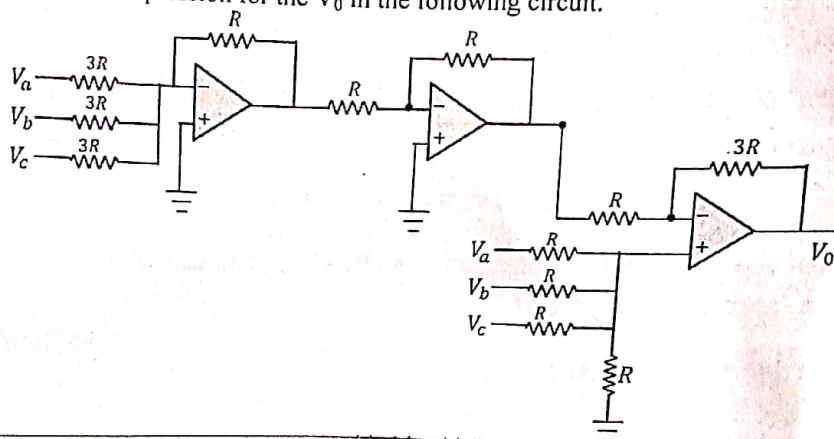
1. A diode operates in the forward bias region with a typical current level I_D . Suppose we wish to increase the current by a factor of 10. How much change in V_D is required? [5]
2. Find the resistivity of (a) intrinsic silicon and (b) p-type silicon with $N_A = 10^{16}/\text{cm}^3$, $n_i = 1.5 \times 10^{10}/\text{cm}^3$, and assume that for intrinsic silicon $\mu_n = 1350 \text{ cm}^2/\text{V.s}$ and $\mu_p = 480 \text{ cm}^2/\text{V.s}$, and for the doped silicon $\mu_n = 1110 \text{ cm}^2/\text{V.s}$ and $\mu_p = 400 \text{ cm}^2/\text{V.s}$. [5]
3. Consider an abrupt p-n junction. Let V_0 be the built-in potential of this junction and V_R be the applied reverse bias. If the junction capacitance C_j is 1 pF for $V_0 + V_R = 1 \text{ V}$, then for what is the value of C_j , when $V_0 + V_R = 4 \text{ V}$. [5]
4. (i) Write the binary equivalent of $(25.625)_{10}$.
(ii) Convert $(35A)_H$ to Octal number system. [5×1]
(iii) Write the smallest integer that can be represented by an 8-bit number in 2's complement form.
(iv) What is the difference between the combinational and sequential circuits?
(v) How to use a XOR gate as a NOT gate?
5. (a) Calculate the base of the number system, where: $\sqrt{224} = 13$. [2]
(b) Design a circuits using minimum number of gates to realize the Boolean function $Y(A,B,C) = \prod(2,6)$. [3]
6. (a) What is the binary equivalent of -2 in an 8 bit signed 2's complement representation? [2]
(b) Design a circuits using minimum number of gates to realize the Boolean function $Z(A,B,C) = \sum(0,4,5)$ [3]
7. (a) Design a Full-Adder using two half adders. [2]
(b) Minimize the Boolean expression $AB \oplus ((A \oplus B).C)$ using Boolean algebra. [3]
8. (a) Design a S-R Latch and write its truth-table. [2]
(b) State the procedure to subtract a number Y from X using 2's complement method. Elaborate with example. [3]
9. In an industry, there are 4 chemical tanks A,B,C and D arranged in the chronological order. A system need to be designed to check and produce a logic '1' output if chemicals in any two consecutive tanks get lower than a predefined threshold level. Consider that the level sensors produce binary logic '1' if chemical in the tank gets lower than the threshold level.
Design the above system using minimum number of logic gates. [5]
10. Derive the expression of rectification efficiency for a full wave rectifier. [5]
11. The load resistance of 'center tapped full wave rectifier' is 500Ω and the maximum value of sinusoidal voltage is 60 volts. Calculate (i) peak, average and r.m.s. value of the current (ii) Ripple factor and (iii) efficiency of rectifier. Consider that the forward resistance and the cut-in voltage of the diode are 50Ω and 0v respectively. [5]
12. Draw and explain the output waveform for circuit given in the following figure. [5]



13. Derive the expression for the output voltage of the following circuit shown in Figure. [5]

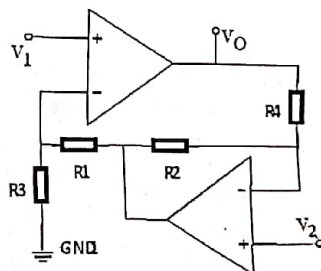


14. Derive the expression for the V_0 in the following circuit. [5]



Group 2

15. (a) Write down the characteristics of an ideal OP-AMP. [3]
 (b) A dual Instrumentation Amplifier is shown in the following figure, If $V_1 = 3\text{ V}$, $V_2 = 5\text{ V}$, $R_1 = R_2 = 20\text{ k}\Omega$ and $R_3 = R_4 = 10\text{ k}\Omega$ then what is the value of output voltage V_0 . [7]



16. What is Pinch-Off in JFET? Draw V-I characteristics of JFET. Also briefly explain about different region of operation. [3+5+2]

17. For a collector feedback circuit the data known are $R_C = 4.7\text{ k}\Omega$, $R_B = 7.75\text{ k}\Omega$. The collector supply voltage and R_C are adjusted to establish a collector current of 1.5 mA at 25°C (Room temperature). Determine the variation of I_C in the temperature range of -65°C to $+175^\circ\text{C}$. [10]

$T(^{\circ}\text{C})$	-65	+25	+175
$I_{CO}(\text{nA})$	1.95×10^{-3}	1.0	33000
β	25	55	100
$V_{BE}(\text{V})$	0.78	0.60	0.225

Important formula:

$$\frac{\Delta I_C}{I_C} = \left(1 + \frac{R_B}{R_C}\right) \frac{M_1 \Delta I_{CO}}{I_C} - \frac{M_1 \Delta V_{BE}}{I_C R_C} + \left(1 + \frac{R_B}{R_C}\right) \frac{M_2 \Delta \beta}{\beta_1 \beta_2}$$

$$\text{where } M_k = \frac{1}{1 + \left(\frac{R_B}{\beta_1 R_C}\right)}$$

18. For a voltage divider biased CE circuit following data is known, $\beta = 100$, $R_C = 2.7\text{ k}\Omega$, $R_E = 1.2\text{ k}\Omega$, $R_2 = 8.2\text{ k}\Omega$, $I_B = 20\text{ }\mu\text{A}$, $V_{CE} = 10.6\text{ V}$ [10]
 The calculate for (i) I_C , (ii) V_E , (iii) V_{CC} , (iv) V_B , (v) R_1

19. (a) Sketch a family of CE transistor output characteristics. Explain the shape of curves qualitatively for (i) Active region, (ii) Saturation Region and (iii) Cut-off region. [8]
 (b) The condition to use approximate analysis for voltage divider biased CE circuit is [2]

Important Equations and constants

Carrier concentration in intrinsic silicon $= \frac{3}{2} e^{-\frac{E_g}{2kT}}$, Junction build in potential $= V_T \ln \left(\frac{N_A N_D}{n_i^2} \right)$, Width of the depletion region

$= \sqrt{\frac{2\epsilon_s}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 + V_R)}$, Charge of an Electron $= 1.6 \times 10^{-19}\text{ coulomb}$, Boltzmann Constant $= 8.62 \times 10^{-5}\text{ eV/K}$, $\epsilon_0 = 11.9$, $B = 7.3 \times 10^{15}\text{ cm}^{-3}\text{K}^{3/2}$, $L_n = 1\text{ }\mu\text{m}$, $L_p = 1\text{ }\mu\text{m}$, $n_i = 1.5 \times 10^{10}\text{ cm}^{-3}$